

Probability Lecture Notes 2: Rules and Conditional Probability

1. Complementary Events

The complement of an event A is the event that A does not happen. It is often written as A'.

$$P(A') = 1 - P(A)$$

Example: If the probability of rain is 0.30, then the probability of no rain is $1 - 0.30 = 0.70$.

2. Addition Rule

For mutually exclusive events, the events cannot happen at the same time.

$$P(A \text{ or } B) = P(A) + P(B)$$

Example: On one die roll, getting a 1 and getting a 6 are mutually exclusive.

$$P(1 \text{ or } 6) = 1/6 + 1/6 = 2/6 = 1/3$$

3. General Addition Rule

If two events can overlap, subtract the probability that both occur so that the overlap is not counted twice.

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

4. Independent Events

Two events are independent when the result of one does not affect the probability of the other.

Examples include tossing a coin twice or rolling two separate dice.

$$P(A \text{ and } B) = P(A) \times P(B)$$

Example: The probability of getting heads on two fair coin tosses is $1/2 \times 1/2 = 1/4$.

5. Dependent Events

Events are dependent when the first event changes the probability of the second.

Example: Drawing two cards from a deck without replacement.

There are 4 aces in 52 cards. After drawing one ace, 3 aces remain in 51 cards.

$$P(\text{two aces}) = (4/52) \times (3/51)$$

6. Conditional Probability

Conditional probability is the probability of event A occurring when we already know that event B has occurred.

$$P(A | B) = P(A \text{ and } B) / P(B)$$

The expression $P(A | B)$ is read as 'the probability of A given B'.

7. Card Example

A standard deck has 52 cards: 26 red cards, 26 black cards, 4 kings, 4 queens, 4 jacks and 4 aces.

Probability of drawing a king:

$$P(\text{King}) = 4/52 = 1/13$$

Probability of drawing a red card:

$$P(\text{Red}) = 26/52 = 1/2$$

8. Quick Review

1. If $P(A) = 0.65$, find $P(A')$.
2. What is the probability of two heads in two fair coin tosses?
3. Why are two card draws without replacement dependent events?